

Course Title	Mathematics-II		
Course Code		Credit	4
Core/ Elective	CEF	Semester	II

Unit –I Linear Algebra:(6 Lectures)

Vector spaces, Subspaces, Linear dependence and independence, Basis and dimension, Dimension theorem. Linear Transformation, Rank – Nulity Theorem (Statement only), Computation of Rank and nullity of LT, Solution of linear simultaneous algebraic equations

Unit –II Eigenvalues and Eigenvectors:(6 Lectures)

Eigenvalues and Eigenvectors, Cayley-Hamilton theorem, Application of Eigen Values and Eigen Vectors: Quadratic form, Diagonalization, Canonical forms and solving system of first order differential equations.

Unit – III Partial Differential Equation: (8 Lectures)

First order PDE, Formation of PDE, Classification of solution: Complete, General and Particular solution, Lagrange's linear PDE, Non-Linear First Order PDE, Some Standard form -I, II, III, IV. Charpit's method. Higher Order Homogeneous linear PDE with constant coefficients, C. F. & P.I, Non-homogeneous PDE with constant coefficients, C. F. & P. I. Method of separation of variables.

Unit- IV Laplace Transform: (7 Lectures)

Laplace transformation and its properties, Unit – step, Impulse and Periodic functions, Error Function. Inverse Laplace Transform, Convolution Theorem, Evaluation of Integral by Laplace Transform, Application of Laplace transform to solution of ODE & PDE.

Unit-V Fourier Series (6 Lectures):

Fourier series, Convergence of Fourier Series, Half range series. Fourier Integral, Fourier sine and Cosine Integral, Complex form of Fourier Integral.

Unit-VI Fourier Transform: (7 Lectures):

Fourier Transform, Fourier Sine and Cosine Transform, Finite sine and cosine transform, Convolution theorem, Application of Fourier Transform to solve boundary value problems (ODE & PDE).

Reference Books:

1. R. K. Jain and S. R. K. Iyenger, Advanced Engineering Mathematics, Narosa Pub. House
2. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
3. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons.
4. QaziZameeruddin & Surjeet Singh, Modern Algebra, S Chand Publication